BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2024

PHYSICS

Past Paper 2024 • Paper Code: PR–IX (A-I-B) (24)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** PR–IX (A-I-B) (24)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. The gravitational force between two bodies having masses 1 kg each and separated by a distance of 1m is:

(A) 6.67×10^{-9}N
(B) 6.67×10^{9}N
(C) 6.67×10^{-11}N
(D) 6.67×10^{11}N

Q2. Kinetic energy is expressed as:

(A) mv2
(B) ½ mv^2
(C) ½ v^2
(D) ½ m^2V

Q3. The percentage efficiency of a nuclear power plant that converts 320 KJ chemical energy to 80 KJ electrical energy is:

(A) 4%
(B) 20%
(C) 25%
(D) 40%

Q4. Young's modulus is measured in unit of:

(A) N/m^2
(B) Kgm
(C) Kgm^{-2}
(D) N/m

Q5. The direction of flow of heat between two objects is determined by their;

(A) Colour
(B) Weights
(C) Temperature difference
(D) Separation

Q6. The transfer of heat from one place to another based on the bulk motion of molecules is carried in:

(A) Radiation
(B) Conduction
(C) Convection
(D) Evaporation

Q7. The SI unit of length among the following is;

(A) Kilogram
(B) Kelvin
(C) Mole
(D) Meter

Q8. In a death well, if a motorcyclist returns to its initial point, the magnitude of his displacement is:

(A) 2πr
(B) 180m
(C) 609m
(D) Zero

Q9. When we push a heavy crate horizontally, the crate does not move due to:

(A) Kinetic friction
(B) Static friction
(C) Sliding friction
(D) Rolling friction

Q10. The centripetal force on a body of mass 3kg moving with velocity 5 m/s in circle of radius 3m is:

(A) 35N
(B) 30N
(C) 25N
(D) 20N

Q11. In a right-angled triangle, the trigonometric function used to find horizontal component of a force is:

(A) sin θ
(B) cos θ
(C) tan θ
(D) cot θ

Q12. A compact disk rotating in a CD player with constant angular velocity is the example of:

(A) Static equilibrium
(B) Translational equilibrium
(C) Rotational equilibrium
(D) Neutral equilibrium

SECTION-B

Time Allowed: 2 Hours 45 Minutes

> Note: Note: Time allowed for Section-B and Section-C is 2 Hours and 45 minutes.

Q2. Attempt any EIGHT parts. Each part carries FOUR marks.

1) Describe thermodynamics and optics by giving one example of each.

2) Define translator motion and describe its three types with examples.

3) Calculate the acceleration of the elevator with a counterweight mass of 1200kg and mass 1100kg when carrying passengers.

4) What is friction? Provide three cases where friction is desirable.

5) In a balanced system, a force of 150N is applied at a distance of 4m from the pivot. Calculate the force needed on the other side at 3m from the pivot.

6) Compare and contrast the concepts of centre of mass and centre of gravity of irregular object highlighting any differences.

7) Obtain the expression for calculating earth's mass using law of universal gravitation.

8) Categorize the following sources of energy as renewable or non-renewable with a brief description of each. (a) Wind energy (b) Fossil fuels

9) Determine the energy produced when a mass of 0.03kg is completely converted into energy.

10) Define atmospheric pressure, compare the atmospheric pressure at sea level and top of Mount Everest.

11) Discuss four factors that affect the rate of heat transfer through a solid conductor.

SECTION-C

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 24

> Note: Note: Attempt any THREE questions. All questions carry equal marks.

Q3.

(a) Elaborate how vectors differ from scalars with two examples.

(b) Specify whether each of the following quantities is a vectors or scalar: Pressure, Momentum, Torque, Work, Electric Current, Temperature

Q4.

(a) Using the law of gravitation, derive formula for orbital speed of satellite.

(b) A satellite orbits the earth 225 km above its surface, find it's orbital speed given that mass of earth is $6x10^{24}$kg and radius of earth is $6.4x10^{6}$mm.

Q5.

(a) Deduce formula for pressure beneath a liquid surface.

(b) The deepest point in an ocean is 12km below sea level, what is the pressure in atmosphere at this point?

Q6.

(a) Explain the molecular process of evaporation at the liquid's surface, providing two examples.

(b) What is the specific heat of a metal substance. If 105 kJ of heat is needed to raise the temperature of 2.3kg of the metal from 16°C to 35.2°C?